at the level of the individual organism. Some biologists go so far as to see DNA as a device used by organisms to reproduce themselves, just as an eye is a device used by organisms to see! Readers of this book will recognize that this attitude is an error of great profundity. It is the truth turned crashingly on its head. They will also recognize that the alternative attitude, the selfish gene view of life, has a deep problem of its own. That problem—almost the reverse one—is why individual organisms exist at all, especially in a form so large and coherently purposeful as to mislead biologists into turning the truth upside down. To solve our problem, we have to begin by purging our minds of old attitudes that covertly take the individual organism for granted; otherwise we shall be begging the question. The instrument with which we shall purge our minds is the idea that I call the extended phenotype. It is to this, and what it means, that I now turn.

The phenotypic effects of a gene are normally seen as all the effects that it has on the body in which it sits. This is the conventional definition. But we shall now see that the phenotypic effects of a gene need to be thought of as *all the effects that it has on the world*. It may be that a gene's effects, as a matter of fact, turn out to be confined to the succession of bodies in which the gene sits. But, if so, it will be just as a matter of fact. It will not be something that ought to be part of our very definition. In all this, remember that the phenotypic effects of a gene are the tools by which it levers itself into the next generation. All that I am going to add is that the tools may reach outside the individual body wall. What might it mean in practice to speak of a gene as having an extended phenotypic effect on the world outside the body in which it sits? Examples that spring to mind are artefacts like beaver dams, bird nests, and caddis houses.

Caddis flies are rather nondescript, drab brown insects, which most of us fail to notice as they fly rather clumsily over rivers. That is when they are adults. But before they emerge as adults they have a rather longer incarnation as larvae walking about the river bottom. And caddis larvae are anything but nondescript. They are among the most remarkable creatures on earth. Using cement of their own manufacture, they skilfully build tubular houses for themselves out of materials that they pick up from the bed of the stream. The house is a mobile home, carried about as the caddis walks, like the shell of a snail or hermit crab except that the animal builds it instead of growing it or finding it. Some species of